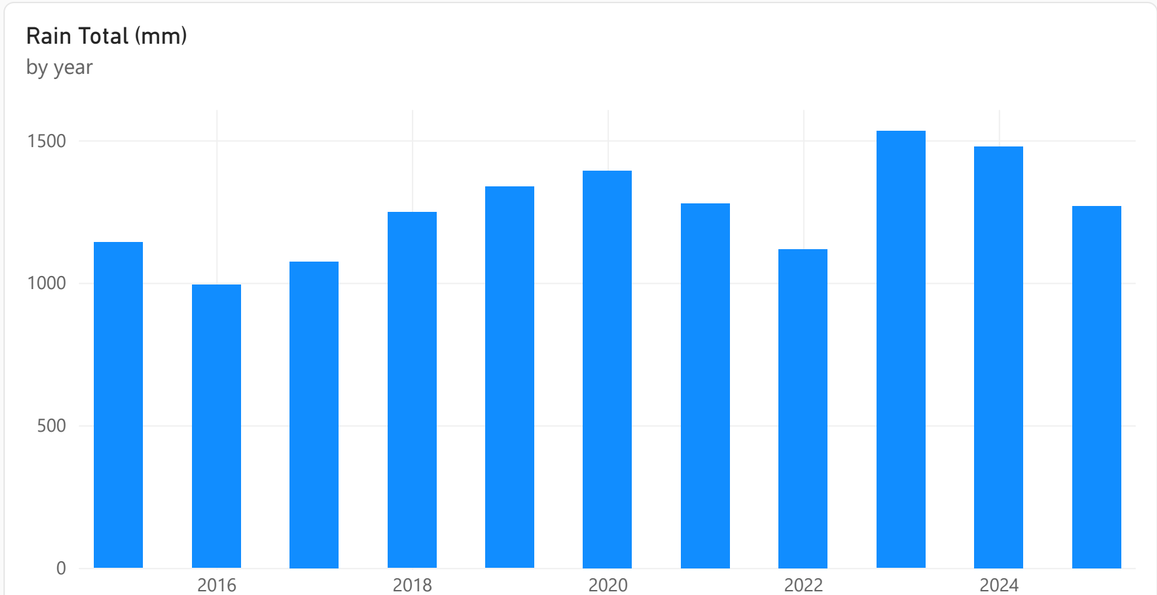
Rain, Rivers and Storm Overflows: Devon and Cornwall

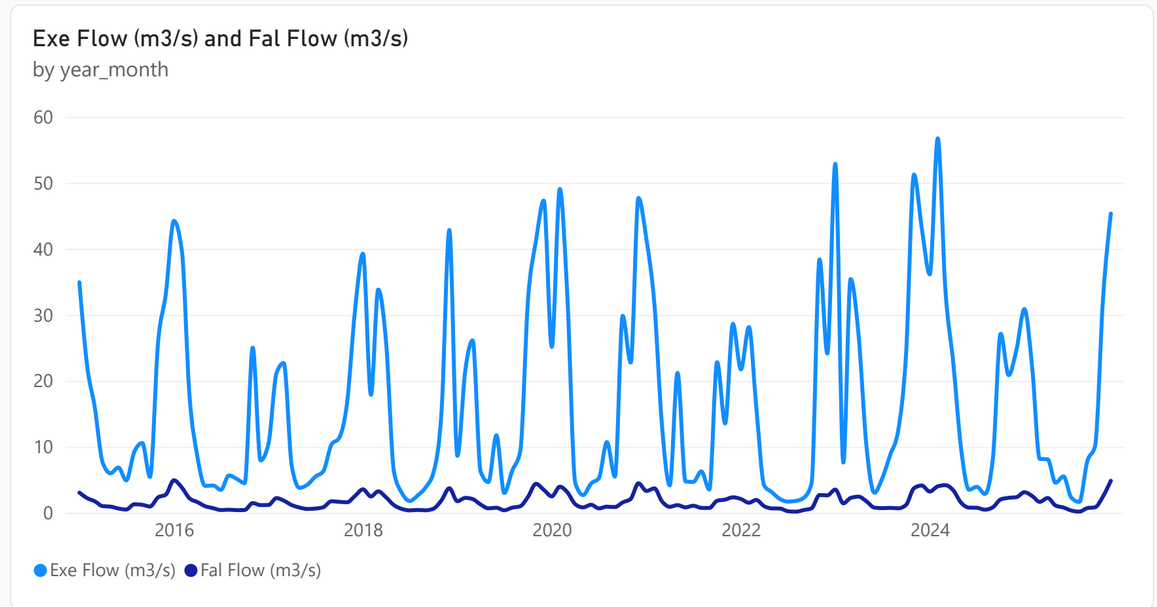
year

2015	2019	2023
2016	2020	2024
2017	2021	2025
2018	2022	

season

Autumn	Summer
Spring	Winter





Rain Total (text)
13,863

Exe Peak Flow (m3/s)
193.6

EDM Events (text)
209,355

Sites With Events (...
1,179

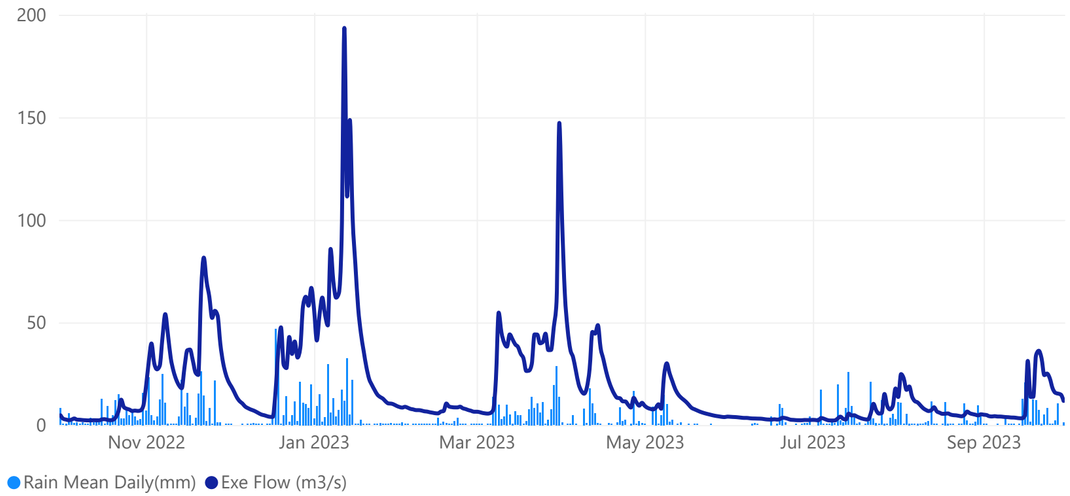
Rain to River

water_year

2023

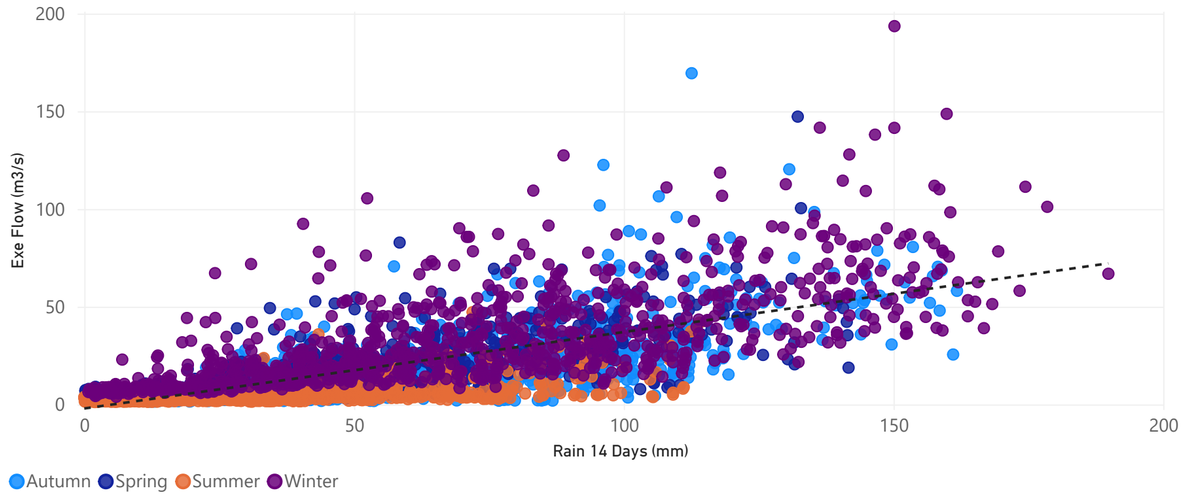
Daily rain and Exe flow

Rain Mean Daily(mm) and Exe Flow (m3/s) by date



Does 14-day rain drive river flow?

Rain 14 Days (mm) and Exe Flow (m3/s) by date and season



Flow rises from about 5 m^3/s after under 20 mm of rain in 14 days to about 50 m^3/s after 100 mm or more.

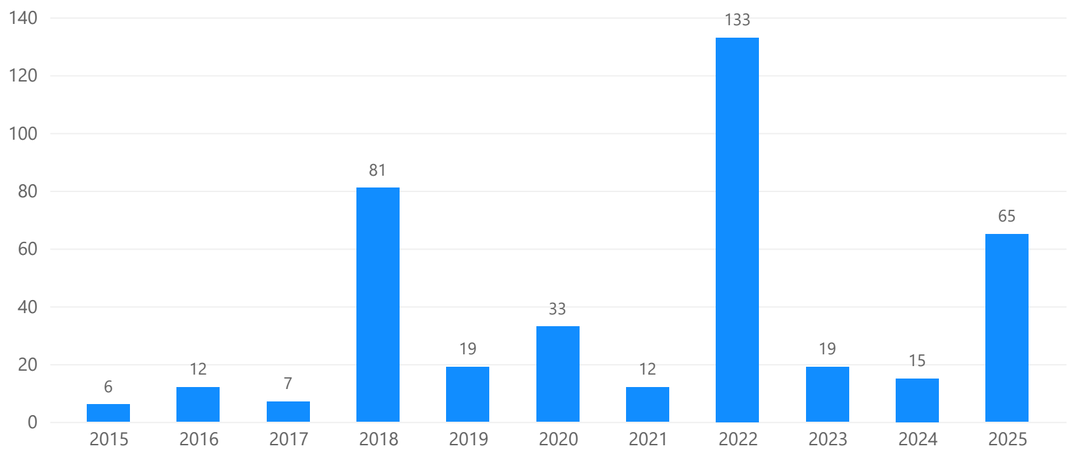
Low Flow

Exe P10 Threshold (m3/s)

2.71

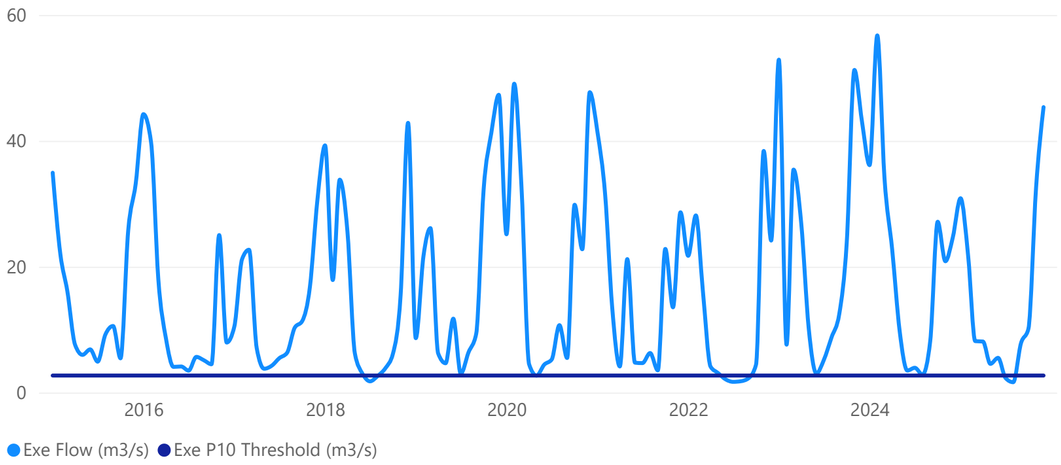
Low Flow Days

by year



Exe flow against its low-flow threshold

Exe Flow (m3/s) and Exe P10 Threshold (m3/s) by year_month



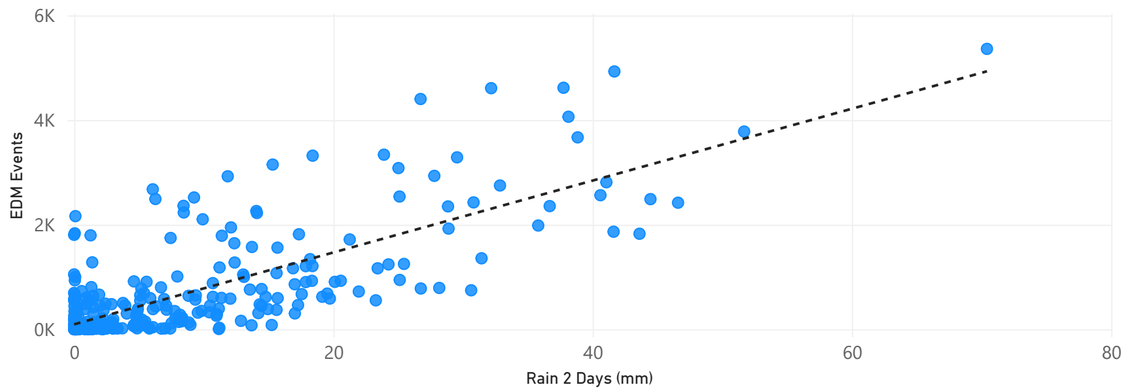
Low-flow days cluster in 2022 (133), 2018 (81) and 2025 (65), against 6–33 in other years. This matters for abstraction and supply planning.

Storm overflows 2025

Sites with events	Overflow events	Overflow hours
1,179	209,355	286,417

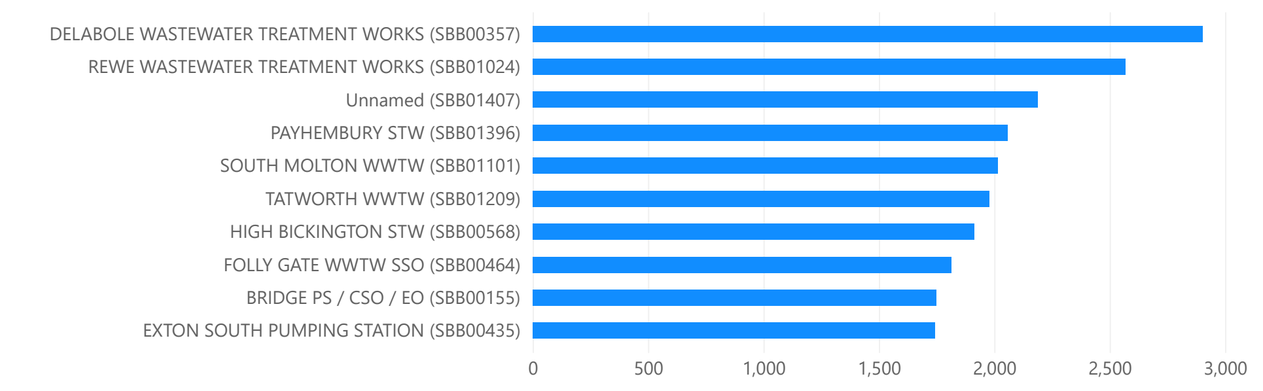
Do storms trigger overflows?

Rain 2 Days (mm) and EDM Events by date



Top 10 sites by overflow hours (events up to 7 days

Sum of hours by site_label



Avg Events Wet Days	Avg Events Dry Days	Wet vs Dry Ratio
1.57K	245.93	6.40

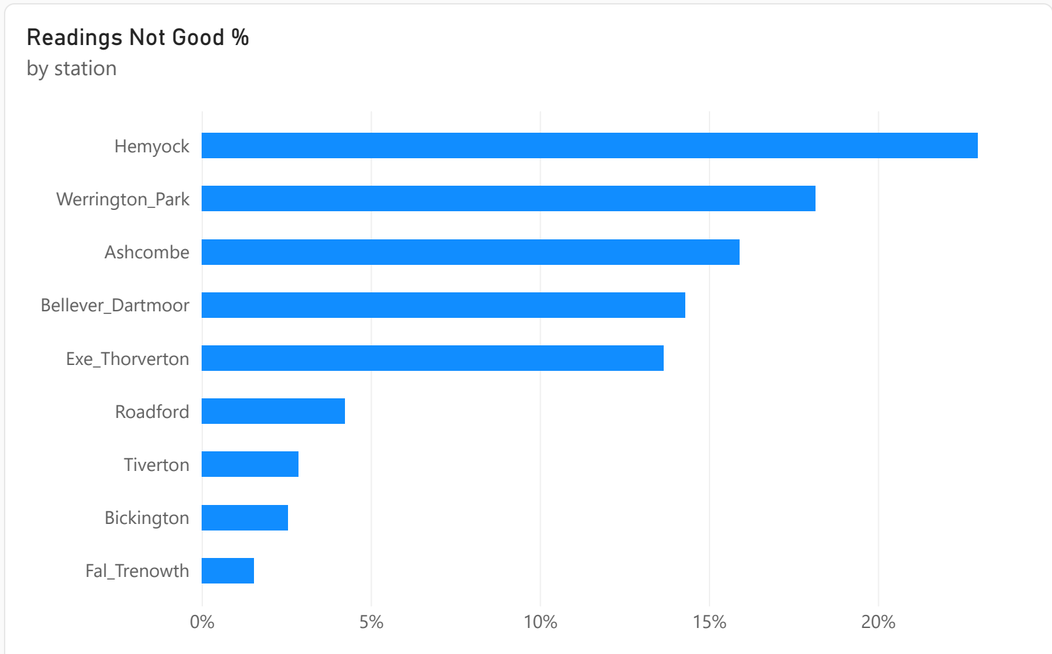
Overflow events are about 6 times more frequent on wet days, and a small group of sites accounts for a large share of discharge hours

Data Quality

Missing Values	Readings Not Good %	EDM Events Excluded (>7 days)
30	10.68%	356

station	Estimated	Good	Missing	Suspect	Unchecked	**Total**
Ashcombe		3379			639	**4018**
Bellever_Dartmoor		3438	6	263	311	**4018**
Bickington		3909	7	102		**4018**
Exe_Thorverton	549	3469				**4018**
Fal_Trenowth	60	3956		2		**4018**
Hemyock		3096		817	105	**4018**
Roadford		3836	12	32	138	**4018**
Tiverton		3898	5	114	1	**4018**
Werrington_Park		3289		529	200	**4018**
Total	**609**	**32270**	**30**	**1859**	**1394**	**36162**

Fal Trenowth had duplicate daily readings for one year; the 09:00 reading was kept. Suspect, Estimated and Unchecked readings are kept but flagged. Events over 7 days are excluded from overflow counts.



Sources
- Environment Agency Hydrology API (rainfall and river flow, 2015–2025)
- South West Water 2025 Detailed EDM Data (storm overflow events)
- Open Government Licence v3

Cleaning
- Fal Trenowth duplicates: kept the 09:00 reading
- Suspect, Estimated and Unchecked readings kept and flagged (10.7%)
- Rain index = mean of 7 gauges, Suspect excluded
- Overflow events longer than 7 days excluded from counts (356 events)
- 23 monitors have no site name in the source file and are shown by their ID

Limitations
- Correlation is not causation
- EDM data is self-reported by the operator and not independently verified; it covers 2025 only
- Event counts are not discharge volumes
- Sites are ranked by monitor ID, so one treatment works can appear more than once

Code and full analysis: github.com/su7ri/water-rainfall-river-overflow